Paper Day-to-Day Learning

- Update probabilities formula (ensure they sum to 1 and are correct)

a) Negative branch ($S_c^t < 0$) (NEGATIVE STIMULUS)

According the paper: $p_c^{t+1} = p_c^t + p_c^t \ell S_c^t$ Eq. 2

$$p_k^{t+1} = \frac{p_k^t - p_k^t p_c^t \ell S_c^t}{1-p_c^t} \quad \text{Eq. 9}$$

Own computation (check if it's true that it adds to unity)

$$\sum_{k\neq c} p_k^{t+1} = \sum_{k\neq c} \frac{p_k^t - p_k^t p_c^t \ell S_c^t}{1-p_c^t} = \sum_{k\neq c} p_k^t - p_k^t p_c^t \ell S_c^t \cdot \left(\frac{1}{1-p_c^t}\right) =$$

$$= \frac{1}{1-p_c^t} \cdot \sum_{k\neq c} p_k^t - p_k^t p_c^t \ell S_c^t = \frac{1}{1-p_c^t} \cdot \sum_{k\neq c} p_k^t (1 - p_c^t \ell S_c^t)$$

$$\boxed{\sum_{k\neq c} p_k^t = 1 - p_c^t}$$

$$= \frac{1 - p_c^t \ell S_c^t}{1-p_c^t} \cdot \sum_{k\neq c} p_k^t$$

$$= \frac{1 - p_c^t \ell S_c^t}{1-p_c^t} \cdot 1 - p_c^t = 1 - p_c^t \ell S_c^t$$

$$p_c^{t+1} + \sum_{k\neq c} p_k^{t+1} = p_c^t + p_c^t \ell S_c^t + 1 - p_c^t \ell S_c^t = \boxed{1 + p_c^t \neq 1}$$

IS WRONG

b) POSITIVE BRANCH ($S_c^t \geq 0$) (POSITIVE STIMULUS)

According the paper: $p_c^{t+1} = p_c^t + (1-p_c^t)\ell S_c^t$ (Eq. 7)

$p_k^{t+1} = p_k^t - p_k^t \ell S_c^t$ (Eq. 9)

OWN COMPUTATION (CHECK IT'S TRUE THAT IT ADDS TO UNITY):

$$\sum_{k\neq c} p_k^{t+1} = \sum_{k\neq c} p_k^t - p_k^t \ell S_c^t = \sum_{k\neq c} p_k^t (1-\ell S_c^t) = 1-\ell S_c^t \sum_{k\neq c} p_k^t =$$

$$= (1-\ell S_c^t)\cdot(1-p_c^t) = 1 - p_c^t - \ell S_c^t + \ell S_c^t p_c^t = \quad \boxed{\sum_{k\neq c} p_k^t = 1 - p_c^t}$$

$$= \left[-p_c^t - \ell S_c^t(1-p_c^t)\right] + 1 = \left(-p_c^t - (1-p_c^t)\ell S_c^t\right) + 1$$

$$p_c^{t+1} + \sum_{k\neq c} p_k^{t+1} = \cancel{p_c^t} + \cancel{(1-p_c^t)\ell S_c^t} + \left(\cancel{-p_c^t} - \cancel{(1-p_c^t)\ell S_c^t}\right) + 1$$

$= \boxed{1}$ IS CORRECT

Paper Day-to-Day Learning

a) Negative branch ($S_c^t < 0$) (NEGATIVE STIMULUS)

HOW IT SHOULD BE:

LET'S DERIVE IT FROM THE FOLLOWING FACTS:

1. The ~~chosen~~ probability of the chosen route is updated by $p_c^{t+1} = p_c^t + p_c^t \ell S_c^t$ Eq. 2

2. The probabilities must sum to 1.

3. ASSUMPTION: Paper says remaining probabilities are updated so that they sum to one, but it never explains how redistribution is performed. Let's assume that every non chosen probability keeps the same proportion relative to the others.

$$\boxed{p_k^{t+1} = r p_k^t \quad k \neq c}$$

Every non-chosen probability is multiplied by same constant r

DERIVATION

$$\boxed{\begin{aligned} 1 - p_c^{t+1} &= \sum_{k \neq c} p_k^{t+1} \\ 1 - p_c^t &= \sum_{k \neq c} p_k^t \end{aligned}}$$

$$\sum_{k \neq c} p_k^{t+1} = \sum_{k \neq c} r p_k^t = r \sum_{k \neq c} p_k^t$$

$$1 - p_c^{t+1} = r(1 - p_c^t) \Rightarrow \boxed{r = \frac{1 - p_c^{t+1}}{1 - p_c^t}}$$

using this expression

$$\boxed{r = \frac{1 - p_c^t - p_c^t \ell S_c^t}{1 - p_c^t}}$$

IF WE COMPARE WITH THE PAPER IT MISSES THIS TERM

$$p_k^{t+1} = p_k^t \left(\frac{1 - p_c^t - p_c^t \ell S_c^t}{1 - p_c^t} \right) = \boxed{\frac{p_k^t - p_k^t p_c^t - p_k^t p_c^t \ell S_c^t}{1 - p_c^t} = p_k^{t+1}}$$

CORRECT VERSION

$$\sum_{k \neq c} \frac{-p_k^t p_c^t}{1-p_c^t} = \sum_{k \neq c} -p_k^t \cdot p_c^t \cdot \frac{1}{1-p_c^t} = \frac{-p_c^t}{1-p_c^t} \sum_{k \neq c} p_k^t = \frac{-p_c^t}{1-p_c^t} \, 1-p_c^t \qquad \boxed{\sum_{k \neq c} p_k^t = 1-p_c^t}$$

$$= -p_c^t$$

↳ That's why paper formula gives $1+p_c^t$, because it misses this term